AICHU TAN

Dublin, CA • thedebbietan@gmail.com
Portfolio: aichutan.github.io | LinkedIn: linkedin.com/in/aichutan

---

PROFESSIONAL SUMMARY

Data Scientist with strong experience in machine learning, deep learning, AI systems, and end-to-end data science workflows. Skilled in building predictive models, time-series forecasting, LLM-powered applications, and computer vision systems. Adept at transforming messy, large-scale datasets into insights and deploying ML solutions using cloud platforms. Background includes public health forecasting, agricultural AI, and full-stack data engineering. Seeking Data Scientist or AI Engineer roles where I can design and deploy impactful, data-driven solutions.

TECHNICAL SKILLS

Programming: Python, SQL, C++, JavaScript
Machine Learning: Scikit-learn, XGBoost, Random Forest, LSTM, Time-Series Forecasting, Feature Engineering
AI & Deep Learning: YOLOv8, LangChain, LLM Integration, NLP, TensorFlow
Data Tools: Pandas, NumPy, RStudio, Tableau, ArcGIS
Cloud Platforms: AWS (EC2, S3, RDS), Azure, Google Cloud, Supabase
Databases: MySQL, BigQuery Sandbox, Azure DB
Deployment: Streamlit, GitHub Pages, Jekyll
Other: MLflow, ERD, Datapane

PROJECTS

End-to-End Dengue Forecasting System | LSTM, XGBoost, Random Forest, Streamlit, Supabase

Created a forecasting platform for weekly dengue cases in Southern Taiwan using 15 years of epidemiological, climate, and mosquito surveillance data.

- Engineered temporal features and trained **tree-based and deep learning models** (LSTM, XGBoost, RF).
- Identified major outbreak drivers such as rainfall, humidity, and lagged case counts.
- Deployed an **interactive Streamlit dashboard** backed by Supabase for real-time exploration.
- Produced an ArcGIS visualization for spatial outbreak monitoring, supporting public health planning.

AI-Powered Plant Disease Detection | YOLOv8, LangChain, LLM, Streamlit, Azure

Designed and deployed a real-time AI system for diagnosing plant diseases from images, supporting smallholder farmers in Africa.

- Trained a multi-class **YOLOv8 computer vision model** for disease detection.
- Integrated a **LangChain-powered LLM pipeline** to generate customized treatment and prevention recommendations.
- Built a **Streamlit web app** connected to an Azure database for structured retrieval of agricultural knowledge.
- Open-sourced and deployed the system on Hugging Face Spaces for public access.

Predicting Restaurant Ratings Using Yelp Metadata | MLflow, XGBoost, RF, Pandas

Built a machine learning pipeline to classify restaurant success (≥4★ vs <4★) using 36,000+ Yelp business records.

- Engineered 61 features and applied **5-fold cross-validation**.
- Tracked experiments using **MLflow** to compare model performance.
- Optimized **XGBoost**, achieving **0.733 accuracy and 0.79 ROC-AUC**.
- Conducted feature importance and clustering to uncover consumer behavior patterns and restaurant archetypes.

Online Ticket Database Management System | MySQL, AWS RDS, SQL, Python

Developed a full-stack ticket reservation system with cloud-hosted relational databases.

- Designed ERD schema and implemented normalized **MySQL** database on **AWS RDS**.
- Built a Python dashboard for ticket analytics and reporting.
- Automated SQL queries for reservations, seat availability, and revenue statistics.

EXPERIENCE

Data Science Researcher - SDSU DiMoLab

Sep 2025 – Present | *San Diego, CA*

- Developed **time-series forecasting models** for dengue case prediction using multi-year health, climate, and mosquito datasets.
- Performed **feature engineering** and outbreak risk modeling.
- Contributed to an AI-driven early-warning framework for public health surveillance.

AI Researcher - Metabolism of Cities Living Lab

Feb 2025 - Present | *San Diego, CA*

- Developed AI tools for indigenous farming communities using **YOLO detection** + **LLM treatment generation**.
- Integrated **Traditional Ecological Knowledge** with machine learning models.
- Co-authored research **accepted at the 4th International One Health Conference (Rome, 2025).**

Business Owner - Wholesale Clothing

Apr 2009 - Oct 2016 | *Taiwan*

- Increased sales by **50%** through data-driven market analysis.
- Optimized supply chain through forecasting and analytics.
- Managed import/export operations across Asia.

Process Engineer - ProMOS Technology

Jul 2004 - Feb 2009 | *Taiwan*

- Improved DRAM yield via **statistical process control (SPC)** and root cause analysis.
- Reduced manufacturing defects through data modeling and process optimization.

EDUCATION

M.S. in Big Data Analytics – San Diego State University – Aug 2024 - Dec 2025
Google Data Analytics Professional Certificate – 2022
B.S. in Chemical Engineering – National Cheng Kung University (Taiwan)